CCIS
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Traffic Light Controller

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Introduction

We designed a traffic light controller using Verilog HDL to control the traffic flow at a road intersection consisting of a main road, side road, and turning lane. The system uses four traffic lights: Main1, Main2, Side, and Turn, each represented by a 3-bit signal indicating red, yellow, or green. The controller operates based on a Finite State Machine (FSM) that transitions between six defined states: Green_Main, Yellow_Main, Green_Turn, Yellow_Turn, Green_Side, and Yellow_Side. Each state controls which lights are active and for how long, using a timer that counts clock pulses to determine when to switch to the next state. This system provides an efficient and organized method for managing traffic at complex intersections using digital logic design.

Description and Explanation

Processes in the Traffic Light Controller

The State Transition Module

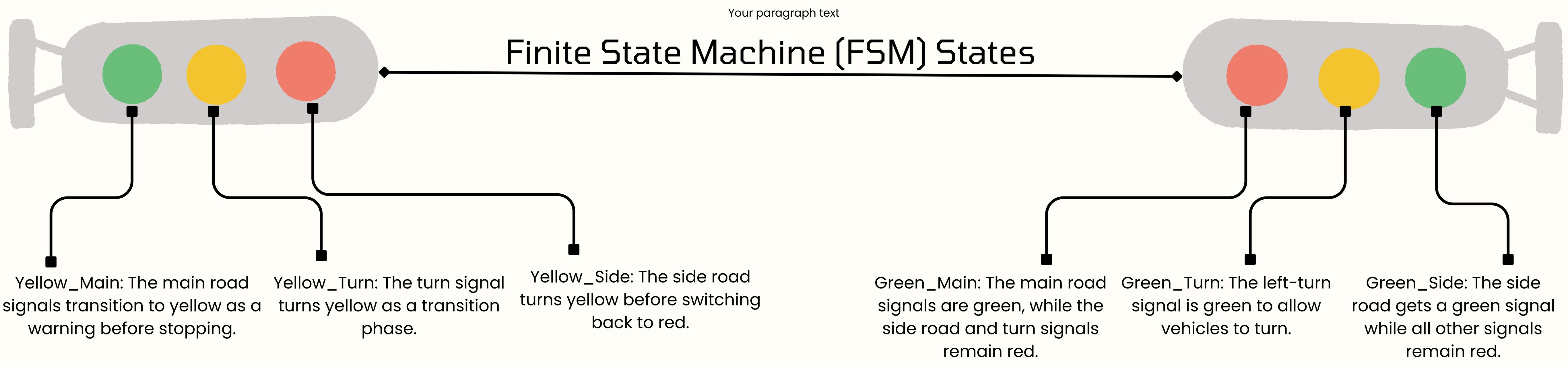
which is responsible for controlling the transition between different traffic light states. It has two inputs: a clock signal and a reset signal. It also has one output, current_state, which represents the active traffic light state.

The Timer Module

which ensures that each state remains active for a predefined duration. The input is current_state, and the output is a timing signal that triggers transitions between states when the predefined time expires.

The Traffic Light Output Module

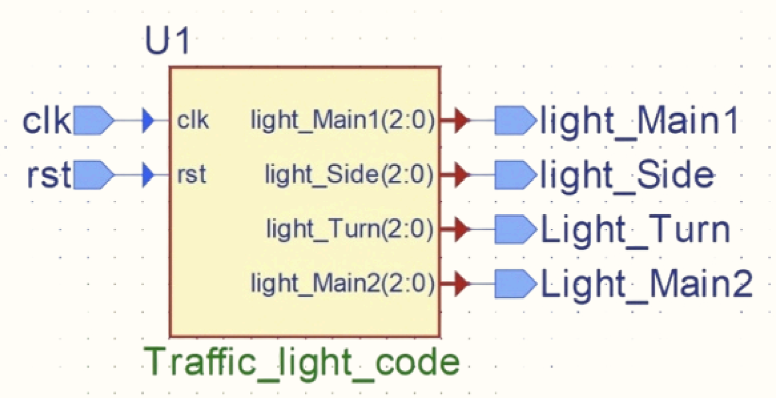
which assigns the appropriate traffic light signals based on current_state. It has one input, current_state, and four outputs: light_Main1, light_Main2, light_Side, and light_Turn. Each output is a 3-bit signal representing red, yellow, and green states



System Operation Scenarios

- When FSM transitions occur, the traffic light signals are updated accordingly. Each state's duration ensures smooth and safe traffic flow.
- When the clock signal (clk) is running, the system transitions between states based on predefined durations. Each state remains active for a specific time before moving to the next
- When the reset signal (rst) is active, the system resets, and the traffic light starts in the Green_Main state, where the main road signals are green.

Diagram



Simulation Trace

Started by testing the states individually with rst = 1 to reset the signal, then set rst = 0 and verify the state transitions. and the tested cases are:

- Rst 1, Clk 1
- Rst 1, Clk 0
- Rst 0, Clk 1
- Rst 0, Clk0

RESET SIGNAL

Make sure the rst signal starts high (1) for a few clock cycles to ensure the system is properly reset.

CLOCK SIGNAL

Ensure that the clk signal toggles regularly (alternating between 0 and 1) and that the frequency matches the specified timer values.

INITIAL STATE TESTING

Start by testing the states individually with rst = 1 to reset the signal, then set rst = 0 and verify the state transitions.